Hydraulic cylinder: from order specification to verified design to acceptance test

Systems engineering dossier. Structure after C. S. Wasson, *System Engineering Analysis, Design, and Development*, 2nd ed. (Wiley, 2016); page numbers are the book's printed pages. All engineering numbers come from `python -m cylver report` (reports in `reports/`, figures in `docs/img/`). Section 12 lists what is assumed. Section 13 lists every source and whether it was checked.

Author: Chander Vinayak Sharma, 9 Nov 2026. Status: analysis and simulation only. No hardware was built or tested.

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0. The question and the answer

Question. A customer orders a custom double-acting cylinder: bore, rod, stroke, pressure, mounting, duty. Which design is verified to be safe and to last, and where are the real failure margins?

Answer in one paragraph. For both example orders, a design sized by the usual static rules (Lamé wall, Euler rod) passes every static check and still fails in service. The failure margins are not where the static rules look:

1. **Fatigue at the tube-to-base weld.** The flat base stops the tube from expanding, which bends the wall next to it. At the original weld position, 10 mm from the base face, this raises the longitudinal stress to 2.2 × the membrane value (HD-160).
    - Rev A fatigue stress factor: **0.38** for the required 4 × 10^5 duty cycles.
    - Moving the girth weld to the point where the edge moment changes sign ($\beta x = \pi/4$, 19 mm from the base face) gives **1.03** with the same tube.
    - The finding was checked with an independent axisymmetric finite-element model of tube and base, with a base free to rotate and no thin-shell assumption. It puts the zero-moment point at 18.9 mm against 19.0 mm from the shell formula, and the edge bending within 1.5 %.
    - At the weld the FE structural stress is 6 % above the shell value, so the design uses the FE value. The shell formula alone would have given 1.09.
    - Thickening the tube instead works poorly. In a design-exploration run (not in the report), a 25 mm wall, 2.2 × thicker, still gave only 0.81 with the weld left at 10 mm.
2. **The cushion chamber is not protected by the port relief valve.** With an overrunning load and cold oil, the cap cushion reached **446 bar** against a 280 bar relief and a 420 bar proof pressure (HD-160 rev A).
    - That peak is also what drives the weld and port fatigue damage.
    - It is fixed at system level: an interface requirement limits the retract command to 0.55 because the load overruns on retraction, and the needle opens to 8 mm².
3. **Buckling of the long loader cylinder (LD-80).** The ordered 50 mm rod reaches first yield at **0.96 ×** the push at relief, once the rod tilt allowed by the bearing clearances, bow, pin eccentricity and self weight are included.
    - The rod-only Euler shortcut says 0.82×; the elastic stepped-column load says 1.28×.
    - A 56 mm rod cannot reach the 1.5× target at any stop-tube length: it peaks at 1.33 near 100 mm and falls again for longer tubes.
    - A 63 mm rod with a 50 mm stop tube reaches **1.65×**.
4. **One order requirement cannot be met as written.** Drift ≤ 5 mm in 10 min (HD-160): oil cooling of only 4.4 K in the locked cap chamber already uses the whole allowance. It goes back to the customer as a specification issue (Wasson Principle 19.4, p. 400: a specification must bound a *feasible* solution space).
5. **Seal friction is calibrated to published measurements**, not assumed (section 7). A first version of this dossier used assumed seal data and concluded that smooth motion at 10 mm/s was infeasible (stick-slip onset 12 mm/s). The measured data put the onset at 1.8 mm/s (PTFE system) and 2.3 mm/s (PU), so R-12 is met. That reversal is the reason to calibrate before concluding.

The verified designs meet every other requirement: HD-160 rev B misses only the drift specification issue, and LD-80 rev C meets all fourteen. They also show **one design method for both orders**:

- move the base weld to the zero-moment point;
- put the ports in the end blocks, not through the tube wall;
- limit the retract command, because the load overruns;
- set each cushion needle for its own end.

LD-80 rev C also takes the PTFE seal system that its own trade study picked in 100 % of weight sets.

The parts are handed to the CNC part-family generator (project 07), which generates and verifies the lathe programs. For LD-80 rev C, 07 accepts and verifies all three parts. For HD-160 rev B, the gland's internal grooves (7.4 mm deep) need a heavier internal grooving bar than 07's base library has (6 mm reach). The bridge adds that bar to the tool library as a tooling requirement, without changing 07, and all three parts then pass.

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1. Need, problem space and concept of operations (Wasson ch. 4-6)

Problem statement (Principle 4.15, p. 91). A make-to-order cylinder maker turns each order line into a design, a set of drawings, NC programs and a test record, often within days. Whether the design is right is usually settled by experience, by catalogue tables and by a final pressure test. The pressure test is ISO 10100 style: leakage, proof, optionally friction (ISO 10100:2020 contents, section 13). That test cannot reveal fatigue, buckling margins or cushion peaks at the customer's duty. Those failures appear in the field, months later.

Stakeholders and their needs (Principle 4.12, p. 89, operational needs):

Stakeholder	Need
Machine builder (customer)	force and speed at the duty; no field failures; known natural frequency for its control
Operator	smooth, predictable motion; load stays where it was put
Maintenance	known wear parts, reseal interval, no surprises
Cylinder maker: design	a repeatable route from order line to verified design
Cylinder maker: production	part-family dimensions that the lathes and their programs can make
Cylinder maker: test bay	pass limits that mean something, traceable to the design

Use cases (Principles 5.12-5.23, pp. 120-122). Each use case has a trigger, a main success scenario and the scenarios that go wrong; the last column is the analysis that covers them.

UC	Title (outcome)	Trigger	Main scenario	When it goes wrong	Covered by
UC-1	Lift the load	operator commands extend	extend at v_max against the load, stop at mid stroke, continue	stall, pressure peaks at the stop	duty simulation
UC-2	Hold the load	valve centred	load stays within the drift limit	seal leakage, oil cooling	drift analysis
UC-3	Lower the load	operator commands retract	controlled retraction	**load overruns**, high speed, cushion peak	duty + cushion simulation
UC-4	Reach the end of stroke	piston enters cushion	decelerate to a gentle stop	over-pressure in the cushion chamber, hard impact	cushion study
UC-5	Position slowly	fine command	smooth slow motion	stick-slip	friction model
UC-6	Accept the cylinder	finished unit on the test bench	leakage, proof, friction tests pass	limits not traceable to the design	ATP (section 9)
UC-7	Manufacture the parts	released design	NC programs generated and verified	part outside tool limits	CNC hand-off (section 10)

Concept of operations (Principle 6.6, p. 139). The order file (`data/orders/*.json`) is the single source of the order (Principle 27.4, p. 579, single information source). `cylver` sizes the tube and base, runs every analysis, derives the requirements status, the FMEA, the acceptance limits and the part-family rows, and writes a calculation report. A

design change is a new revision in the same order file, with its reason. The report shows the revision history. Each revision is a decision with its evidence.

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2. Phases, modes and states (Wasson ch. 7)

Wasson separates *phases* of the life cycle (Section 7.3, p. 150), *modes* the user selects (Principle 7.4, p. 157) and *states* (operational, environmental and dynamic; Principles 7.3, 7.8 and 7.9, pp. 156-166). Each mode and state that loads the cylinder becomes a **load case**: modes and states are where the load cases come from.

Kind	Name	What it does to the cylinder	Load case in `cylver`
phase	manufacture	weld, machine, assemble	weld detail category, CNC hand-off
phase	acceptance test	proof pressure, leakage, friction	proof stress check, ATP
phase	operation	the duty cycle	duty simulation
phase	maintenance	reseal; clearances grow with wear	clearance tilt (buckling), extrusion gap
mode	extend	cap side pressurised, push	buckling at full extension; duty
mode	hold	both ports blocked	drift; oil cooling
mode	retract	rod side pressurised; load may overrun	cushion at the cap end; duty
mode	float / off	no pressure	(not a load case)
dynamic state	cushioning	trapped oil decelerates the mass	cushion study, 40 °C and 0 °C oil
dynamic state	stop at mid stroke	valve closes in 50 ms	duty pressure peaks
dynamic state	stall at relief	pressure at the relief setting	static checks at relief pressure
environmental state	cold oil (0 °C)	viscosity about 10 × higher	cushion peak, friction
environmental state	aerated oil	effective bulk modulus drops	natural frequency

Allowable and prohibited actions (Principle 7.10, p. 169). The analysis produced one prohibited action that only the machine can enforce: *full-speed retraction under an overrunning load*. It becomes the interface requirement IR-1 in section 4.

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3. Requirements derived from the order and the standards (Wasson ch. 19-22)

How they are derived (Section 21.4, p. 436). Every requirement comes from one of three places:

- an order clause (force, speed, pressure, duty, drift, smooth speed);
- a standard's method (ISO 10100 acceptance modules, the ISO/TS 13725 approach to buckling, EN 1993-1-9 weld fatigue, ISO 6022 / ISO 6020-2 mounting interfaces; ISO/TS 13725 applies to cylinders that conform to ISO 6020, ISO 6022 and ISO 10762);
- a derived need found by analysis, for example "the cushion chamber is not protected by the relief, so its peak needs its own requirement".

Each requirement follows Principle 22.5 (p. 449): subject, *shall*, action, outcome, bound. Each names its verification method (Principle 22.4, p. 448). The methods are Wasson's Inspection, Examination, Analysis, Demonstration and Test (Principle 13.10, p. 280). Where the order is silent on conditions (oil temperature, the thermal state during holding), Principle 19.20 (p. 407) on environmental conditions applies and the gap is reported.

ID	Requirement (short)	Source	Method	HD-160 rev B	LD-80 rev C
R-01	extend at ≥ 0.95 v_max against the load	order	A; T	met	met
R-02	tube von Mises ≤ Re/2 at relief (Pm)	Lamé; design rule	A	met	met
R-03	no yield at proof pressure	ISO 10100 §8.3 (factor assumed 1.5)	A; T	met	met

ID	Requirement (short)	Source	Method	HD-160 rev B	LD-80 rev C
R-04	base plate ≤ 1.5 Re/2 (Pb, conservative bound)	plate theory	A	met	met
R-05	base junction ≤ 2 Re (PL+Pb+Q, shakedown)	ASME VIII-2 Part 5 categories	A	met	met
R-06	rod thread ≤ Re/2 at relief pull	ISO 898-1	A	met	met
R-07	first yield at full extension ≥ 1.5 × push at relief	ISO/TS 13725 approach; 1.5 = project target	A	met (1.65)	met (1.65)
R-08	every fatigue location survives the required duty cycles	order duty; S-N sources	A (axisymmetric FE at the weld); T (ISO 10771-1)	met (1.03, weld)	met (1.10, weld)
R-09	piston-seal extrusion gap ≤ allowable	seal makers' gap charts (values assumed, tables not retrieved)	A; I	met	met
R-10	cushion peak ≤ proof pressure, 0-40 °C	derived (relief does not protect the cushion)	A; T	met (324 / 420 bar)	met (273 / 330 bar)
R-11	end-stop impact ≤ 0.10 m/s	project target	A; T	met (0.080)	met (0.090)
R-12	no stick-slip down to v_min	order; seal data from Sârbu et al. 2022/2024	A; T (ISO 10100 module F)	met (onset 1.8 mm/s)	met (onset 1.8 mm/s)
R-13	drift ≤ 5 mm / 10 min	order	A; T (ISO 10100 module P)	**not met** (4.4 K cooling uses it up)	met only below 5.9 K cooling
R-14	lowest natural frequency reported	TPM (Section 21.8, p. 442)	A	11.0 Hz	11.1 Hz
R-15	mounting interface per ISO 6022 (25 MPa series) where the order names an ISO mounting	ISO 6022 / ISO 6020-2	I	**open**: tables in the paid standard	**open**

The full statements, results and limits for every revision are in `reports/HD-160.md` and `reports/LD-80.md`. Traceability runs both ways (Principle 19.13, p. 402; Section 21.7, p. 439):

- each requirement names its source;
- the report gives the analysis that verified it;
- the acceptance procedure (section 9) names the requirement each test limit protects.

Specification issues found (Principle 19.4, p. 400, feasibility):

- **R-12 (withdrawn as an issue).** With assumed seal data the model put the stick-slip onset at 12 mm/s, above the 10 mm/s requirement. Calibrated to measured friction (section 7), the onset is 1.8 mm/s, so the requirement is met. It stays verified by test (F-2), because seal friction varies between batches and with wear.
- **R-13.** Drift is defined without a thermal condition. A locked 32 L cap chamber that cools by 4.4 K in 10 minutes uses the whole 5 mm allowance (oil expansion coefficient assumed 7 × 10^-4 /K). The order must say "at constant oil temperature" or give the cooling rate.

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4. Architecture and interfaces (Wasson ch. 26-27)

Physical architecture (Principle 26.5, p. 551):

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rear pin — base (welded) — tube — head/gland (bearing, rod seal, wiper) — rod — rod eye — front pin
               |                              |
          cap cushion                 rod cushion spear
          port A (tube or block)      port B (tube or block)
        piston (guide rings, piston seal) on rod thread + nut
```

Interfaces (Principles 27.1, p. 575; 27.6, p. 583). Each interface has an owner (Principle 27.3, p. 577) and becomes either a requirement or a load case.

ID	Interface	Across it	Owner	Becomes
IF-1	rear and front pins	axial force; eccentricity ≤ 1 mm; no moment	machine builder	buckling eccentricity, pin sizes
IF-2	ports A/B ↔ hoses ↔ valve	flow, pressure; hose compliance	machine builder	dynamic model; port location
IF-3	port relief valves	relief setting	machine builder	static checks at relief; *not* the cushion
IF-4	valve command	command profile	machine builder (control)	**IR-1: the retract command shall be limited so the overrunning load does not exceed v_max (0.55 for HD-160, 0.6 for LD-80)**
IF-5	oil	temperature 0-70 °C, air content	machine builder	viscosity, bulk modulus
IF-6	design → production	part-family rows	cylinder maker	CNC hand-off (section 10)
IF-7	design → test bay	pass limits	cylinder maker	ATP (section 9)

IF-3 and IF-4 are where the HD-160 rev A cushion failure came from: the relief valve on the machine side does not protect a volume inside the cylinder. FMEA is applied to interfaces as well as parts (Principle 27.9, p. 596).

Software architecture of the verification tool (`cylver/`). One order file goes in. Each module owns one physics question, and all of them share the `Design` object. The dynamic model is the one source of pressure histories, so cushion peaks, fatigue spectra, natural frequency and stick-slip all come from the same equations.

Module	Question	Feeds
`model.py`	geometry, sizing of tube and base, materials	all
`stress.py`	Lamé, plate, edge bending, thread	requirements, fatigue
`buckling.py`	stepped beam-column FE, P_cr, first yield with imperfections	R-07, FMEA
`dynamics.py`	valve + hoses + chambers + cushions + friction + stops	duty history, cushion, frequency, loop gain
`fatigue.py`	rainflow, S-N, Miner, equal-damage test pressure	R-08, ATP Q-1
`seals.py`	Stribeck friction, stick-slip, drift	R-12, R-13, ATP F/P
`requirements.py`, `fmea.py`, `acceptance.py`, `aoa.py`	systems-engineering layers	report
`cnc_bridge.py`	part-family rows → project 07	manufacturing

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5. Budgets and margins (Wasson ch. 31)

A design safety margin is "an additive or multiplicative factor that is applied to a Measure of Performance… to accommodate variations and uncertainty" (Section 31.2, p. 667). Every capability needs a budget and a margin (Principle 31.1, p. 667). The margins used here, and where each came from:

Margin	Value	Basis
static primary membrane	Re / 2	project design rule (common practice); ASME VIII-2 categories for the split
static primary bending	1.5 Re / 2	rectangular-section shape factor 1.5
primary + secondary (junction)	2 Re	shakedown (ASME VIII-2 Part 5), from memory
buckling	first yield ≥ 1.5 × push at **relief**	project target, applied to a *usable* load, not to Euler
fatigue, welds	γ_Mf = 1.35 on stress range	EN 1993-1-9 safe-life, high consequence

Margin	Value	Basis
fatigue, machined	j = 1.5 on stress	FKM order of magnitude
cushion	peak ≤ proof pressure	derived: the proof test is the highest pressure the unit has seen without damage

Pressure budget. Supply ≤ relief < proof, and the cushion peak must stay ≤ proof. HD-160: 250 / 280 / 420 bar, with the cushion peak 446 bar in rev A and 324 bar in rev B.

Drift budget, apportioned (Principle 31.4, p. 668): 5 mm = thermal contraction + piston-seal leakage over 10 min. The thermal share is computed and the remainder is the leak budget, which becomes the ISO 10100 module P pass limit. For HD-160 the remainder is zero: that is how R-13 was found.

Reliability index for buckling (stress-strength interference, Principle 34.21, p. 752). Re CoV 7 % and load CoV 10 % (both assumed) give β = 4.25 for HD-160 rev B, about 1 × 10^-5 per extreme push.

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6. Analysis of alternatives (Wasson ch. 32)

Method after Wasson Fig. 32.7 (p. 690):

1. Criteria with weights.
2. Utility functions (linear, 0-10).
3. Two to four viable candidates.
4. Scoring.
5. Ranking.
6. **Sensitivity**: each weight ±10 % (Wasson's definition, p. 682) and 2000 random weight sets.

Each score is marked **model** (computed by `cylver`) or **judgement** (rationale written next to it). An undocumented trade study "is nothing more than personal opinion" (Principle 32.5, p. 697), so the judgement scores stay visible and editable in `cylver/aoa.py`.

Trade	Result (HD-160 rev B)	Robust?	What decides it
Seal system	PTFE step seal + PU tandem, PTFE piston seal	HD-160: the PU system is **not viable** (extrusion margin 0.90 < 1, R-09), so it is screened out before scoring (Wasson's viable-alternative rule, p. 683). LD-80: PTFE wins 81 % of weight sets; no ±10 % weight change flips it	**model (measured-data seals):** stick-slip onset 1.8 vs 2.3 mm/s, so both are smooth at 10 mm/s; lower friction and a larger extrusion margin for PTFE. These outweigh the judgement criteria (dry rod, contamination, cost) that favour PU.
Rod surface	HVOF WC-CoCr (65 % of weight sets) ahead of laser cladding (24 %); hard chrome last (6 %)	no ±10 % flip, but **all scores are judgement**, so the ranking is only as good as the scores	the regulatory criterion: Cr(VI) restriction dossier submitted 11 Apr 2025, opinions pending (news source, section 13). Coupon tests (salt spray ISO 9227, abrasion) would replace the judgement scores.
Tube material	E355 +SR	wins 100 % of weight sets	**model:** the stronger 25CrMo4 tube is sized thinner (7.5 mm instead of 11.25 mm), so its *weld* fatigue factor drops from 1.03 to 0.54 (LD-80: 1.10 to 0.80). Higher strength does not help a weld.

The tube trade is the useful lesson. A static-strength argument picks the stronger steel. The fatigue model picks the stock tube, because the governing location is a weld and weld S-N curves do not improve with parent-metal strength (EN 1993-1-9 detail categories are independent of grade).

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7. Modelling and simulation (Wasson ch. 33)

"All models are wrong but some are useful" (Principle 33.5, p. 718). The fidelity of each model is stated (Principle 33.1, p. 706), and each was **verified** against an independent result. None is **validated** against hardware yet. The

acceptance tests in section 9 are designed to do that.

Model	Fidelity	Verified against (test in `tests/test_cylver.py`)	Not modelled
Lamé, plates	closed form, linear elastic	thin-wall limit; radial equilibrium	plasticity, autofrettage
base-junction edge bending	thin-shell beam-on-elastic-foundation, rigid base	closed-form edge moment; zero at βx = π/4	base rotation; rm/t 4-9 is near the theory's limit (both checked by the FE row below)
base junction, continuum	axisymmetric FE (`axisym.py`): Q4 elements, 2x2 Gauss, tube wall 8 elements thick, base plate meshed, about 1500 elements; through-wall linearised stress	membrane far from the base equals the thick-wall value exactly; edge bending within 1.5 % of the shell formula; zero-moment point 18.9 vs 19.0 mm (HD-160) and 8.8 vs 8.9 mm (LD-80); at the chosen welds 5-6 % above the shell value, which the fatigue check now uses	weld geometry and residual stress (covered by the weld detail category); the sharp corner itself (singular, not used)
buckling	Euler-Bernoulli FE, 160 elements, geometric stiffness, P-δ with imperfections	Euler for 4 end conditions (0.2 %); exact two-segment equation (0.2 %); secant formula and 1/(1-P/Pcr) amplification (1 %)	the ISO/TS 13725 equations themselves; shear; spherical-bearing friction; rod-in-tube contact beyond the two bearings
rainflow / Miner	ASTM E1049 three-point	the standard's worked example, exact counts	sequence effects; mean stress for welds (deliberately ignored, as in EN 1993-1-9)
dynamic model	lumped: 2 chambers + 2 lines + valve lag, orifice edges, cushions, Stribeck friction, stiff stops	natural frequency = oil-column formula (2 %); loop-gain limit converges to Merritt's K_v < 2ζω_h as the valve gets fast (to 0.90-0.95 ×, the remainder from the asymmetric areas); cushion monotonic in needle size	line dynamics (waves), pump flow limit, valve spool forces, temperature rise, cavitation details, dissolved-air release
bulk modulus	Merritt series form, free air by gas law	liquid limit; air softens at low pressure	air dissolving at high pressure (Gholizadeh et al. 2011): conservative
seal friction	Stribeck per seal element (Coulomb level growing with pressure, static excess, Stribeck speed, viscous); tandem rod seal = two elements	**calibrated** to Sârbu et al. (Polymers 2024, Tables 3 and 5; 60 mm PTFE and PU coaxial seals, 0-160 bar). Predicted stick-slip onset 0.52 mm/s (PTFE) and 2.3 mm/s (PU) against the measured 0.2-0.3 and 1.4 mm/s (Polymers 2022): conservative by about 2 ×, consistent with the damping the model leaves out	the viscous coefficient (assumed; matters only at speed); PU coaxial data used for a PU U-cup; batch scatter
stick-slip	1-DOF, constant-flow drive, Stribeck	linear criterion dF/dv < 0 agrees with the simulation (stick-slip at 0.5 × onset, smooth at 2 × onset)	frictional memory and lag

Two model-based findings that only the simulation could give:

1. **Overrunning retraction.** With a critical-centre valve, the overrunning load sets the retraction speed:

    - HD-160 retracted at 0.41 m/s against a 0.20 m/s order speed.
    - LD-80 retracted at 0.52 m/s against 0.30 m/s.

    That speed is what overloads the cap cushion: 446 and 463 bar with 0 °C oil. Limiting the command (IR-1) brings the entry speed to 0.21 and 0.28 m/s and the peaks to 324 and 273 bar. 2. **Natural frequency and damping.** The oil-column frequency at mid stroke is 11.7 Hz (HD-160) and 12.5 Hz (LD-80 rev C). The formula and the model's eigenvalue agree to 0.1 Hz. The lowest values over the stroke are 11.0 and 11.1 Hz. - With a critical-centre valve and tight seals, the damping ratio at null is only 0.0014 (HD-160) and 0.0028 (LD-80). A position loop could then use a velocity gain of only 0.2-0.4 1/s (Merritt's 2ζω_h, which the model reproduces for a fast valve). - With the 30 ms proportional valve assumed here, the model allows 1.2-2.5 1/s, because the slow valve attenuates the loop at the resonance. That is a fragile kind of stability, not a design margin. - Anyone closing a loop around this cylinder needs added damping: a bleed orifice, pressure feedback, or a valve with underlap. This goes to the machine builder as a TPM (R-14).

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8. Reliability and FMEA (Wasson ch. 34)

FMEA with RPN = S × O × D (Wasson eq. 34.41, p. 765, after Dhillon). Occurrence is **read from the computed margin** with a fixed rule (`cylver/fmea.py`), so the ranking moves when the design changes. Severity and detection are judgement, written per row. Ranking for HD-160 rev A, the "as ordered" design:

Failure mode	Margin	O	RPN
tube-to-base weld fatigue	0.38	9	630
port cross-bore fatigue (ports in the tube wall)	0.33	9	441
drift under load	0.89	9	315
cushion over-pressure	0.94	9	288
column buckling	1.65	3	180
piston-nut thread fatigue	6.5	2	160
piston-seal extrusion	1.41	5	150

For rev B, the occurrence drops for three rows. The weld goes from O = 9 to 7 (margin 1.03), the port from 9 to 2 (margin 3.25) and the cushion from 9 to 5 (margin 1.30). The top of the rev B ranking is:

Failure mode	Margin	O	RPN
base weld fatigue	1.03	7	490
drift	0.89	9	315
buckling	1.65	3	180
cushion over-pressure	1.30	5	160

The weld stays at the top because its margin is thin, and that margin depends on an inspected root (FAT 71). With an uninspected root (FAT 36, IIW) the same weld would fail. **Weld-root inspection is therefore a derived manufacturing requirement**, not a quality nicety.

Other reliability points:

- The strength-stress view (Principle 34.21, p. 752) is applied to buckling (β above).
- Wear-out (Principle 34.20, p. 751) enters through the clearances: the buckling tilt and the extrusion gap both grow with wear. The inspection step I-1 records them so that a reseal can restore the analysed state.

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9. Verification and validation (Wasson ch. 13 and 28)

Verification asks "did we build it right?" and validation asks "did we build the right thing?" (Principles 13.3 and 13.4, p. 275). The verification methods are listed per requirement (section 3). The acceptance test procedure below applies three principles:

- one test verifies as many requirements as practical (Principle 28.3, p. 608);
- the test equipment is calibrated (Principle 28.9, p. 618);
- **every pass limit is a number the design analysis relied on**, so a test that passes also validates the model.

Step	Module	Test	Pass limit (HD-160 rev B; values in the report)	Protects
I-1	inspection	piston and gland bearing clearances	≤ design values	R-07 (tilt), R-09 (gap)
L-1	ISO 10100 §8.2	leakage at low test pressure	no external leakage	standard module
L-2	ISO 10100 §8.3	proof / external leakage	no leakage, no permanent set; analysis: 314 MPa < 450 MPa at 420 bar	R-03
P-1	ISO 10100 §9	piston-seal leakage	leak budget from R-13 (zero for HD-160: open item)	R-13

Step	Module	Test	Pass limit (HD-160 rev B; values in the report)	Protects
F-1	ISO 10100 §10.6.2	static friction	≤ 1.3 × model breakaway	R-01, R-12
F-2	ISO 10100 §10.4.2	friction at constant speed 0.1-100 mm/s, fit the Stribeck model	stick-slip onset of the fitted curve ≤ 10 mm/s	R-12; validates the seal model
C-1	project test	cushion at 0 °C and 40 °C, transducer **in the cushion chamber**	≤ proof; impact ≤ 0.10 m/s	R-10, R-11
Q-1	ISO 10771-1 Annex B (type test)	cyclic pressure 0 → p_test	p_test = equal-damage pressure for 10^6 cycles (report)	R-08

What is and is not checked in the standards. ISO 10100's contents (modules, clause numbers) were checked. Its test pressures and durations were not available, so the procedure says "per ISO 10100" instead of inventing them.

The ISO 10771-1 scope says the user sets the test pressure, so this project computes it: the constant-amplitude pressure that does the same Miner damage at the governing location as the design life.

- For HD-160 rev B this is **183 bar for 10^6 cycles**. For LD-80 rev C it is **150 bar**.
- Both are *below* the rated pressure, because most of the duty damage comes from a few peaks above the S-N knee and the rest falls under the cut-off.
- So a type test at rated pressure for 10^6 cycles is more severe than the design life at this location. Passing it shows the life with margin; failing it would not by itself prove the design wrong.
- For the rev A designs the equal-damage pressure is undefined. The port peak exceeds static strength with the elastic Kt, and the report says "n/a, redesign first" instead of printing a number.

The evaluator (`acceptance.evaluate`) fits the Stribeck model to module F data and judges F-1 and F-2. It is demonstrated on **synthetic** data (model + noise), clearly labelled; it shows the code works, not that a cylinder passes. A test with a "sticky" seal set (static/Coulomb × 3) is caught by F-2.

Integration is incremental (Principle 28.1, p. 605): inspection first, then pressure tests, then function tests, then the type test. A failure is isolated by the step it appears in (Principle 28.2, p. 606).

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10. Manufacturing hand-off (the CNC part-family generator)

The verified design produces the rows that project 07 (`cnc-part-family-generator`) consumes: rod end, piston and gland. `cylver/cnc_bridge.py` imports 07 unchanged, generates the FANUC canned-cycle program and runs 07's material-removal verifier on it. The part-family table is derived from the design, so a design revision regenerates the NC programs too.

Part	HD-160 rev B	LD-80 rev C
rod end	PASS	PASS
piston	PASS	PASS
gland	PASS with a heavy internal grooving bar (10 mm reach; grooves 7.4 mm deep)	PASS

The first run rejected the HD-160 gland: 07's base tool library reaches 6 mm with its internal grooving bar. That is a tooling requirement derived from the design, not a part to redesign. `data/cnc_tooling.json` adds the heavier bar, and the bridge uses it only for parts that need it; 07 itself is unchanged.

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11. Results summary

	HD-160 rev A (as ordered)	HD-160 rev B	LD-80 rev A (as ordered)	LD-80 rev B	LD-80 rev C
tube OD / rod (mm)	182.5 / 100	182.5 / 100	90 / 50	90 / 63	90 / 63

	HD-160 rev A (as ordered)	HD-160 rev B	LD-80 rev A (as ordered)	LD-80 rev B	LD-80 rev C
buckling (first yield / push at relief)	1.65	1.65	**0.96**	1.65	1.65
governing fatigue stress factor	**0.33** (port in tube)	1.03 (weld, FE)	**0.35** (port in tube)	**0.39** (port in tube)	1.10 (weld, FE)
cushion peak (bar), 0-40 °C	**446**	324	**463**	**411**	273
requirements not met	R-08, R-10, R-13	R-13	R-07, R-08, R-10	R-08, R-10, R-11	none
open (needs a paid standard)	R-15	R-15	R-15	R-15	R-15
CNC hand-off (project 07)	-	3 / 3 PASS (gland with heavy grooving bar)	-	-	3 / 3 PASS

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12. Limits and assumptions (what is assumed vs sourced)

- **Not validated on hardware.** Every result is analysis and simulation. The acceptance procedure is the plan to validate the models.
- **Seal friction is calibrated to one published test series** (two 60 mm coaxial seals, PTFE and PU, 0-160 bar, Sârbu et al. 2022/2024). The viscous coefficients are still assumed, and the PU U-cup uses PU coaxial-seal data. **The extrusion-gap limits are still assumed:** the seal makers' tables could not be retrieved. Module F on the real cylinder would replace all of these.
- **ISO/TS 13725 equations were not available.** The buckling model follows the standard's published principles (geometric model, off-axis load, own weight, fully extended) with a classical beam-column FE. The first-yield criterion and the 1.5 target are this project's choices.
- **Material values are from memory or secondary summaries** (marked `checked: false` in `data/materials.json`). E355 +SR uses 450 / 580 MPa from a mill catalogue summary; a recalled 520 / 640 was not confirmed and is not used.
- **Base junction:** the shell solution is now checked by an axisymmetric FE model, which agrees on the zero-moment point to 0.1 mm. Both are linear elastic and homogeneous. Weld geometry and residual stress are covered only by the weld detail category, so strain gauges near the weld on the type-test unit remain the hardware check (OPEN_ITEMS O-7).
- **The port Kt of 3** is a typical cross-bore value (assumed). Ports in end blocks are modelled as Kt × pressure. Peaks above Rm with the elastic Kt are reported as static failures, not given a fatigue life.
- **The duty cycle is a design duty cycle** (extend with a mid-stroke stop, hold, retract, hold) invented for the example. The real spectrum must come from the customer or from measurement.
- **The pump is an ideal pressure source.** Line wave dynamics are not modelled, and air is treated as free (conservative at high pressure).
- **Proof pressure factor 1.5** is assumed until checked in ISO 10100 (a vendor guide, S25, gives the same typical value; the ISO text is paid and was not available).
- **The groove sizes in the CNC rows are illustrative**, not seal-catalogue values.

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13. References and how far each was checked

"Checked" means the cited text was read in this project (page or section stated). "Not re-checked" means the formula or value is quoted from memory or from a search summary. Every not-re-checked item is covered by a test against an independent closed form, or is listed in section 12 as an assumption.

#	Source	Used for	Status
W	C. S. Wasson, *System Engineering Analysis, Design, and Development*, 2nd ed., Wiley 2016	dossier structure; the principles cited per section (book page numbers)	checked: chapters 4-7, 13, 19-22, 26-28, 31-34 extracted and the cited principles read
S1	ISO 10100:2020 *Hydraulic fluid power - Cylinders - Acceptance tests*	ATP structure: modules L (8.2, 8.3), P (9), F (10.4.1, 10.4.2, 10.6.2)	table of contents checked (DIN preview); test pressures and durations NOT available
S2	ISO 10771-1:2015 *Fatigue pressure testing of metal pressure-containing envelopes*	type test Q-1	scope checked via store pages: test pressure set by the user, ISO/TR 10771-2 rating method, Annex B = cylinders; full text not available
S3	ISO/TS 13725:2021 *Method for evaluating the buckling load of a hydraulic cylinder*	buckling principles	scope checked via store pages: geometric model, off-axis load, own weight, fully extended; based on Hoblit and NFPA T3.6.37; not for internal rod buckling; spherical-bearing friction not included. The equations of the standard were NOT available and are not reproduced
S4	H. Gholizadeh, R. Burton, G. Schoenau, "Fluid bulk modulus: a literature survey", Int. J. Fluid Power 12(3), 2011, pp. 5-15	effective bulk modulus, entrained vs dissolved air, critical pressure	checked (paper read)
S5	H. E. Merritt, *Hydraulic Control Systems*, Wiley 1967	chamber continuity, hydraulic natural frequency, K_v < 2 zeta omega_h	not re-checked; natural frequency and gain rule reproduced by tests against the model's own eigenvalues
S6	M. Jelali, A. Kroll, *Hydraulic Servo-systems*, Springer 2003	valve-cylinder modelling	not re-checked
S7	R. G. Budynas, J. K. Nisbett, *Shigley's Mechanical Engineering Design*, "pressurized cylinders"	Lame	not re-checked; test: thin-wall limit and radial equilibrium
S8	S. Timoshenko, S. Woinowsky-Krieger, *Theory of Plates and Shells*, 2nd ed.	circular plate; cylinder edge bending (beam on elastic foundation)	not re-checked; test: edge moment closed form and zero-moment point
S9	S. Timoshenko, J. Gere, *Theory of Elastic Stability*	stepped bars, secant formula, amplification 1/(1-P/Pcr)	not re-checked; tests: FE vs Euler (4 end conditions), vs the exact two-segment equation, vs secant formula and amplification
S10	ASTM E1049-85 *Standard Practices for Cycle Counting in Fatigue Analysis*	rainflow	not re-checked; test: the standard's worked example (counts 3:0.5, 4:1.5, 6:0.5, 8:1, 9:0.5)
S11	EN 1993-1-9 *Fatigue*	weld S-N shape, gamma_Mf	not re-checked
S12	IIW recommendations (Hobbacher, 2nd ed. 2016)	FAT 71 / 36 for one-sided butt welds with / without root inspection	not re-checked
S13	FKM guideline *Analytical strength assessment*	machined steel endurance order of magnitude	not re-checked
S14	VDI 2230 Part 1 (2015)	thread endurance sigma_ASV, load factor phi_n	not re-checked
S15	ISO 898-1	thread stress area	test against the tabulated M20 value (245 mm^2)
S16	B. Armstrong-Helouvry, P. Dupont, C. Canudas de Wit, "A survey of models, analysis tools and compensation methods for the control of machines with friction", Automatica 30(7), 1994	Stribeck model, stick-slip	not re-checked; test: simulation agrees with the linear criterion

#	Source	Used for	Status
S17	H. K. Mueller, B. S. Nau, *Fluid Sealing Technology*	pressure-dependent seal friction, extrusion	not re-checked
S23	F. A. Sârbu, F. Arnăuţ, A. Deaconescu, T. Deaconescu, "Theoretical and experimental research concerning the friction forces developed in hydraulic cylinder coaxial sealing systems made from polymers", *Polymers* 16(1):157, 2024, doi:10.3390/polym16010157 (CC BY)	seal friction vs pressure and speed (Tables 3, 5)	checked: full text read, tables used
S24	same group, "Experimental research on polymer-based coaxial sealing systems of hydraulic cylinders for small displacement velocities", *Polymers* 14(2):290, 2022, doi:10.3390/polym14020290 (CC BY)	seal size (60 × 4 mm); measured stick-slip onsets (PTFE < 0.2 mm/s, PU < 1.4 mm/s)	checked: full text read
S25	hydraulic-cylinders.net, "Hydraulic cylinder pressure testing: ISO standards and quality control procedures" (vendor guide)	proof pressure "typically 1.5 × the maximum working pressure"	secondary, vendor source; ISO 10100 §8.3 itself not available
S18	EN 10305-1 (E355 +SR tube), EN 10083-2/-3 (C45, 42CrMo4, 25CrMo4)	material values	not re-checked; E355 +SR 450/580 MPa from a mill catalogue seen as a search summary (my recalled 520/640 was NOT confirmed and is not used)
S19	ECHA Cr(VI) restriction process: Annex XV dossier 11 Apr 2025; consultation 18 Jun - 18 Dec 2025; RAC/SEAC opinions pending (Freyr regulatory news, 25 May 2026)	coating trade, regulatory criterion	checked via search summary of the news item; final adoption date unknown
S20	ASME BPVC Section VIII Division 2, Part 5 (design by analysis)	stress categories Pm / Pb / PL+Pb+Q and the 2 Re shakedown limit	not re-checked
S21	ASTM D341	viscosity-temperature (Walther)	not re-checked; test: VG 46 = 46 cSt at 40 C by construction
S22	ISO 3320	preferred bores and rods	not re-checked (values only used as a list)

14. Decision log

Each decision names the evidence it rests on (Wasson Principle 32.5, p. 697: an undocumented trade study is only opinion). "Report" means `reports/HD-160.md` or `reports/LD-80.md`.

#	Decision	Alternatives considered	Evidence
D-01	Buckling criterion: first yield with clearance tilt, bows, eccentricity and self weight ≥ 1.5 × the push at **relief** pressure	safety factor on rod-only Euler; on the elastic stepped-column load	Euler is not consistently conservative: LD-80 rev A Euler 0.82 ×, elastic 1.28 ×, first yield 0.96 × (report)
D-02	LD-80: rod 63 mm + 50 mm stop tube	rod 56 with any stop tube; lower relief setting	rod 56 peaks at 1.33 × near a 100 mm stop tube; 63 + 50 gives 1.65 × (exploration runs, rev B in the report)
D-03	Girth weld at the zero-moment point of the base edge bending (19 mm HD-160, 9 mm LD-80)	thicker tube; stronger tube	HD-160 weld 0.38 → 1.03; a 25 mm wall with the weld at 10 mm gave only 0.81; 25CrMo4 makes it worse (0.54)
D-04	Weld fatigue uses the axisymmetric FE structural stress	thin-shell formula with a rigid base	FE is the better model (flexible base, real thickness) and is 5-6 % higher at the chosen welds; the conservative choice
D-05	Ports in the base and head blocks	ports through the tube wall	port fatigue 0.33 → 3.25 (HD-160)
D-06	Interface requirement IR-1: retract command limited (0.55 HD-160, 0.6 LD-80)	stronger cushion parts; counterbalance valve (outside the cylinder scope)	the overrunning load retracts at 0.41 / 0.52 m/s and drives the cap cushion to 446 / 463 bar
D-07	Separate cushion needle per end (LD-80 rod end 0.6 mm^2)	one needle size for both ends	the 63 mm rod leaves a small annulus; a shared 3 mm^2 needle let the piston hit the

#	Decision	Alternatives considered	Evidence
			stop at 0.19 m/s
D-08	PTFE step seal + PU tandem rod seal, PTFE piston seal	PU U-cup + PU compact	HD-160: PU not viable (extrusion margin 0.90 < 1); LD-80: PTFE wins 81 % of weight sets
D-09	Seal friction calibrated to published measurements; assumed values retired	keep assumed values until the bench test	assumed data gave a false "R-12 infeasible"; the calibrated model reproduces measured onsets within 2 × (test)
D-10	E355 +SR tube	25CrMo4 +QT	weld fatigue decides, not static strength (section 6)
D-11	F-2 pass limit = stick-slip onset of the fitted bench curve ≤ v_min	a limit on the static/Coulomb friction ratio	with measured data the onset depends on the Stribeck speed more than on the static excess; a ratio limit came out non-binding
D-12	Heavy internal grooving bar added to the tool library (`data/cnc_tooling.json`)	smaller seal cross-section for the large gland	the seal cross-section follows the rod size; the tool is a shop input; 07 stays unchanged
D-13	ISO 6022 (25 MPa series) as the mounting reference for both orders	ISO 6020-2 (16 MPa compact)	both orders are rated 200-250 bar, above the 16 MPa series; the dimension tables themselves are open (O-4)
D-14	Proof factor 1.5 × max(rated, relief), assumed	none available without the standard	ISO 10100 §8.3 value not available; vendor guide gives 1.5 as typical (S25); open item O-1

What remains open needs a paid standard, a supplier, a customer answer or hardware: see OPEN_ITEMS.md.